

Ashitha Paramesha Gowda

ashithapgowdaapg@gmail.com (667) 481-1679 linkedin.com/in/ashitha-p-gowda/ ashithapgowda.com

EXPERIENCE

Graduate Research Assistant (BSPH, JHU – Baltimore, MD)

Mar 2025 – Present

- Led technical requirement gathering with public health teams to define data extraction needs from Medicaid waiver PDFs across 15+ U.S. states.
- Developed a rule-based parser using Python, PyMuPDF, and regex to extract structured data from varied document layouts, reducing manual processing effort by 80%.
- Prototyped a bounding-box-based detection module using Google Document AI to improve extraction accuracy on complex, layout-heavy PDFs.

Software Developer - Full Stack (JKTech, Bengaluru)

May 2023 – July 2024

- Engineered folder-level file linking across S3, Box, and OneDrive using recursive indexing and pre-signed URLs in a microservices-based environment, reducing manual file mapping by 30%.
- Diagnosed and resolved a critical blockchain sync bug where the system misreported readiness; introduced validation checks and logged new failure events in the event-driven pipeline to prevent data loss and enhance transparency.
- Operated within Agile sprints to author test suites for event-driven flows and collaborate with QA on fallback plans, increasing test coverage by 40% and reducing regression cycles by 25%.
- Resolved 80% of backend tickets by tracing failures using Kibana logs and replaying EventStore event chains, reducing average triage time by 2x.

Software Developer - Backend (Juego Studios, Bengaluru)

Jan 2022 – Apr 2023

- Contributed to 14+ commercial projects including card games, betting platforms, and educational multiplayer apps; responsibilities included feature dev, bug fixes, and debugging legacy codebases.
- Designed and built a plug-and-play communication module for NJS2 framework supporting 1:1 and group messaging with voice via WebRTC; shipped with built-in room management and player moderation controls.
- Developed a secure device tracker app with LokiJS and SSO integration; implemented CLI flags (`--local`, `--admin`) to bypass SSO for internal testing and uncovered a critical server-side security flaw, reducing admin bottlenecks by 70% and prompting a core infrastructure patch.

Software Developer - Intern (Juego Studios, Bengaluru)

Aug 2021 – Dec 2021

- Built a real-time multiplayer Snake & Ladder game from scratch using Node.js, Socket.io, and Argon; supported six-player rooms with latency-aware reconnection logic.
- Developed reusable backend handlers for client disconnects and rejoin flows, reducing session drop errors by 50%.
- Collaborated with senior developers to implement modular session architecture and track performance bottlenecks using test environments.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, SQL, Java
- **Frameworks & Libraries:** Node.js, NestJS, Express, Django, Flask, Jest, Mocha, Chai
- **Databases:** PostgreSQL, MongoDB
- **Cloud & DevOps:** AWS, GCP, Docker, Kubernetes, Jenkins, GitHub Actions, CI/CD
- **Messaging & Streaming:** Kafka, RabbitMQ, AWS SQS/SNS
- **Tools & Collaboration:** Git, Postman, REST API, TDD, Jira, Confluence

EDUCATION

Johns Hopkins University

Master's in Computer Science, GPA 3.5/4.0

Baltimore, MD

Aug 2023 – May 2025

Vidyavardhaka College of Engineering

Bachelor of Engineering in Information Science, GPA 3.8/4.0

Karnataka, India

Aug 2017 – May 2021

PERSONAL PROJECTS

Founder - Full Stack Developer (CNotes.in – CRM Platform, Bengaluru)

Jan 2021 – Present

- Built and deployed a cable TV CRM platform using PHP, MySQL, HTML/CSS, and Textlocal API, serving over 30,000 subscribers across 50+ operators by developing billing, SMS receipts, and payment tracking workflows tailored for last-mile collection needs; improving on-time payments by 35%.
- Collected operator feedback through direct calls and field visits, integrating high-priority UX updates that led to a 90%+ customer retention rate.

Personal Portfolio Website

Dec 2024

- Developed using Next.js and MUI to serve as a centralized showcase of technical projects and resume.
- Designed *XOXO AI Game*, a humanized variant of Tic-Tac-Toe that blends the Minimax algorithm with randomized behavioral choices to simulate emotion-driven moves and enhance engagement; fully playable in portfolio project page.